CAGAYAN KURT JOHN B.
2CS - A

SMARTPHONE. JAVA

```java
package phones;
import java.util.ArrayList;
import java.util.List;
// Smartphone class with nested App class
public class Smartphone {
  // Class fields
  private String brand;
  private String model;
  private double storageCapacity;
  private List<App> apps;
  private boolean isPoweredOn;
  // Constructor to initialize the smartphone with pre-installed apps
  public Smartphone(String brand, String model, double storageCapacity) {
      this.brand = brand;
      this.model = model;
      this.storageCapacity = storageCapacity;
      this.apps = new ArrayList<>();
      this.isPoweredOn = false;
      // Pre-installed apps
      apps.add(new App("Gmail", "2025.03.23.744886800"));
      apps.add(new App("Google Maps", "25.14.00.742022593"));
      apps.add(new App("Chrome", "135.0.7049.79"));
  }
  // Method to turn the smartphone on
  public void turnOn() {
      if (!isPoweredOn) {
          isPoweredOn = true;
          System.out.println("The smartphone is now powered on.");
      } else {
          System.out.println("The smartphone is already powered on.");
      }
  }
  // Method to turn the smartphone off
  public void turnOff() {
      if (isPoweredOn) {
          isPoweredOn = false;
          System.out.println("The smartphone is now powered off.");
      } else {
          System.out.println("The smartphone is already powered off.");
      }
  }
  // Method to install an app (only works when the smartphone is on)
  public void installApp(String appName, String version) {
```

```java
    if (isPoweredOn) {
        App app = new App(appName, version);
        apps.add(app);
        System.out.println(appName + " has been installed on the smartphone.");
    } else {
        System.out.println("Cannot install app. The smartphone is off.");
    }
}
// Method to uninstall an app (only works when the smartphone is on)
public void uninstallApp(String appName) {
    if (isPoweredOn) {
        if (apps.isEmpty()) {
            System.out.println("No apps to uninstall.");
            return;
        }
        for (int i = 0; i < apps.size(); i++) {
            if (apps.get(i).getAppName().equals(appName)) {
                apps.remove(i);
                System.out.println(appName + " has been uninstalled from the smartphone.");
                return;
            }
        }
        System.out.println("App " + appName + " not found on the smartphone.");
    } else {
        System.out.println("Cannot uninstall app. The smartphone is off.");
    }
}
// Method to display all installed apps (only works when the smartphone is on)
public void displayInstalledApps() {
    if (isPoweredOn) {
        if (apps.isEmpty()) {
            System.out.println("No apps installed.");
            return;
        }
        System.out.println("Installed apps:");
        for (App app : apps) {
            System.out.println(app);
        }
    } else {
        System.out.println("Cannot display apps. The smartphone is off.");
    }
}
// Nested class representing an App (Composition inside Smartphone)
private class App {
    private String appName;
    private String version;
    // Constructor to initialize app details
    public App(String appName, String version) {
        this.appName = appName;
```

```java
        this.version = version;
    }
    // Getter for appName
    public String getAppName() {
        return appName;
    }
    // Method to get app details
    @Override
    public String toString() {
        return "App: " + appName + ", Version: " + version;
    }
  }
  // Getters for new fields
  public String getBrand() {
      return brand;
  }
  public String getModel() {
      return model;
  }
  public double getStorageCapacity() {
      return storageCapacity;
  }
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    }
    // Getter for appName
    public String getAppName() {
        return appName;
    }
    // Method to get app details
    @Override
    public String toString() {
        return "App: " + appName + ", Version: " + version;
    }
}
// Getters for new fields
public String getBrand() {
    return brand;
}
public String getModel() {
    return model;
}
public double getStorageCapacity() {
    return storageCapacity;
}
}
```

```java
package phones;
public class App {
    private String appName;
    private double version; // e.g., 1.0, 1.1
    private double storageReq; // In GB
    private Smartphone smartphone; // Bi-directional reference to Smartphone
```

```java
    // Constructor
    public App(String appName, double version, double storageReq) {
        this.appName = appName;
        this.version = version;
        this.storageReq = storageReq;
    }
    // Getters
    public String getAppName() {
        return appName;
    }
    public double getVersion() {
        return version;
    }
    public double getStorageReq() {
        return storageReq;
    }
    public Smartphone getSmartphone() {
        return smartphone;
    }
    // Setters
    public void setSmartphone(Smartphone smartphone) {
        this.smartphone = smartphone;
    }
    @Override
    public String toString() {
        return "App: " + appName + ", Version: " + version + ", Storage: " + storageReq + " GB";
    }
}
```